Fourier series

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IBVP for Vibrating string with no external forces

- We consider the one-dimensional wave equation problem in a computational domain

$$(x, t) \in [0, L] \times [0, \infty).$$

- The IBVP under consideration consists of the following:

- The governing equation:

$$u_{tt} = c^2 u_{xx}, (x, t) \in [0, L] \times [0, \infty). \tag{1}$$

  - The boundary conditions for all $t > 0$:

$$u(0, t) = 0, u(L, t) = 0. \tag{2}$$

  - The initial conditions for $0 \leq x \leq L$ are

$$u(x, 0) = \phi(x), u_t(x, 0) = \psi(x). \tag{3}$$

HISTORICAL REMARKS on Fourier Series

- The theory of Fourier series has its historical origin in the middle of the eighteenth century, when several mathematicians were studying the vibrations of stretched strings and some other problems involving steady-state heat conduction.

- For the case of a string stretched between the points $x = 0$ and $x = \pi$, Daniel Bernoulli (in 1753) gave the solution of (1) as a series of the form

$$u(x, t) = b_1 \sin x \cos ct + b_2 \sin 2x \cos 2ct + \cdots \tag{4}$$

  - Observe that:

    - A typical term of this series, $b_n \sin nx \cos nct$, is a solution of (1).

    - Further, every finite sum of such terms is a solution.

    - The series (4) will also be a solution if term-by-term differentiation of the series is justified.

HISTORICAL REMARKS on Fourier Series (contd.)

- When t = 0, the series (4) reduces to

$$u(x,0) = b_1 \sin x + b_2 \sin 2x + \cdots \tag{5}$$

- This should give the initial shape of the string, that is, the curve

$$u = u(x,0) = \phi(x).$$

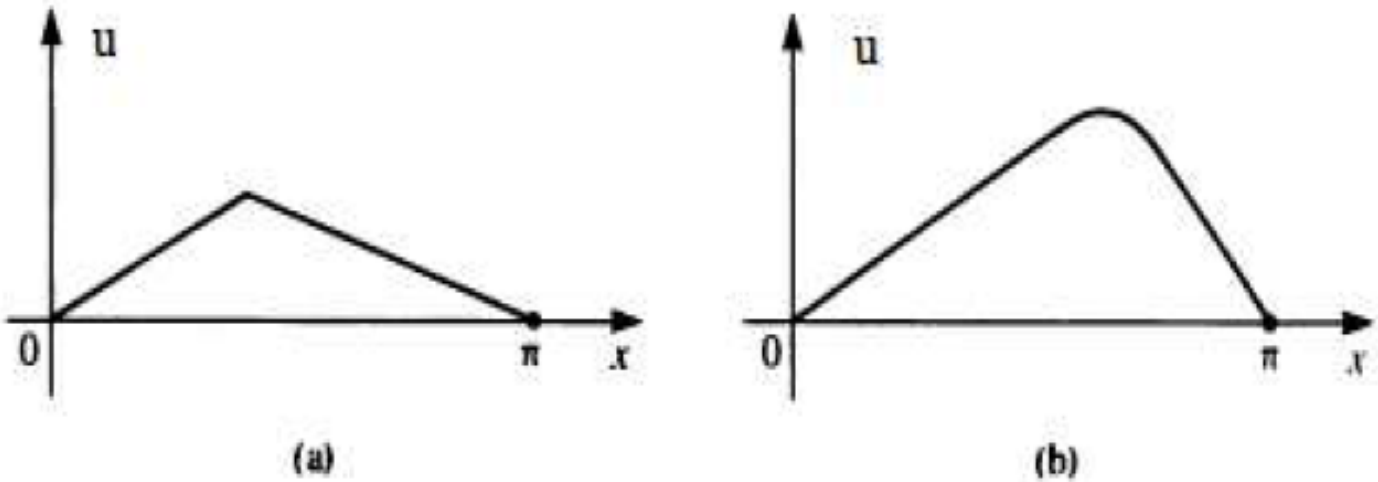
- Thus, we should have

$$\phi(x) = b_1 \sin x + b_2 \sin 2x + \cdots \tag{6}$$

- It is clear on physical grounds that there is a great amount of freedom in the way the string can be constrained in its initial position.

HISTORICAL REMARKS on Fourier Series (contd.)

- For Example:
  - (a) If the string is plucked aside at a single point, then the shape will be a broken line,
  - (b) If it is pushed aside by using a circular object of some kind, then the shape will be partly a straight line, partly an arc of a circle, and partly another straight line.



- It is reasonable to ask whether the single analytic expression (6) could represent a straight line on part of the interval $[0, \pi]$, a circle on another part, and a second straight line on still another part?

Fourier Series on *its way*

- Therefore, as a result of mathematically analyzing this physical problem, Bernoulli arrived at an idea that has had very far-reaching influence on the history of mathematics and physical science, namely, the possibility that any function can be expanded in a trigonometric series of the form (6).

- However, D'Alembert and Euler rejected Bernoulli's idea - questioning the authenticity of such an (absurd) idea.

- The controversy bubbled on for many years, and in the absence of mathematical proofs, no one could convince anyone else to his way of thinking.

Fourier Series on its way

- In 1807, the French physicist-mathematician Fourier announced in this connection that an arbitrary function f(x) can be represented in the form (6).

- He supplied no proofs, but instead heaped up the evidence of many solved problems and many convincing specific expansions - in fact, so many that the mathematicians of the time began to spend more effort on proving, rather than disproving, his conjecture.

- The first major result of this shift in the winds of opinion was the classical paper of Dirichlet in 1829, in which he proved with full mathematical rigour that such a series actually exists.

Let us know Fourier



Jean Baptiste Joseph Fourier (March 21, 1768 – May 16, 1830) was a French mathematician and physicist who is best known for initiating the investigation of Fourier series and their application to problems of heat flow. He is also known for Fourier transform and Fourier's heat law.

Some Facts:

Alma Mater: Ecole Normale Suprieure

Academic Advisor: Joseph-Louis Lagrange

Notable students:

Peter Gustav Lejeune Dirichlet

Claude-Louis Navier

Giovanni Plana

Definition (Periodic functions)

A function $f(x)$ is called a periodic function if $f(x)$ is defined for all real x, and if there is some positive number T, called a period of $f(x)$, such that

$$f(x+T) = f(x).$$

If there is a smallest positive constant T with this property, it is called the fundamental period (also primitive period or basic period or prime period).

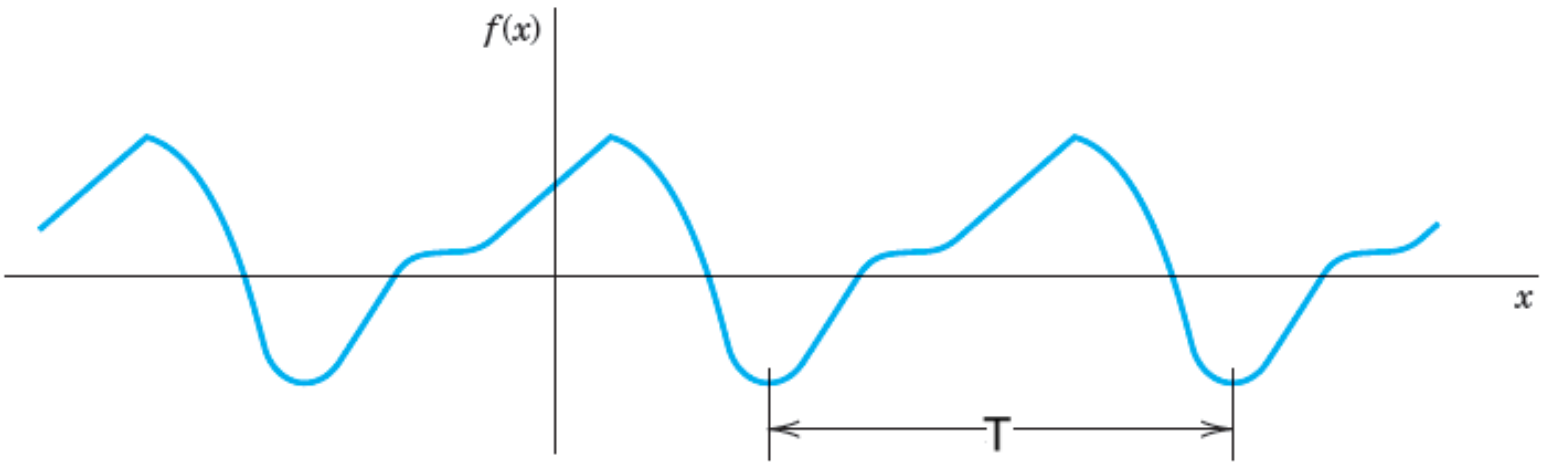


Figure 1: Periodic function of period T

Remark

- If $f(x)$ has period T, it also has the period $2T$ because $f(x+2T) = f[(x+T)+T] = f(x+T) = f(x)$, etc.; thus for any integer $n = 1, 2, 3 \ldots$,

$$f(x+nT) = f(x)$$

for all x.

- A function with period T will repeat on intervals of length T and these intervals are referred to as periods.

- Geometrically, a periodic function can be defined as a function whose graph exhibits translation symmetry. Specifically, a function $f(x)$ is periodic with period T if the graph of f is invariant under the translation in the x-direction by a distance of T.

Example

1. The sine function is a periodic function with period 2π since

$$\sin(x + 2\pi) = \sin x$$

2. A function on the real numbers that gives the fractional part of its argument is an example of periodic function with period 1. Eg

$$f(1.2) = f(2.2) = f(3.2) = \cdots = 0.2$$

3. $f(x) = e^{ikx}$ is a periodic function with period $\frac{2\pi}{k}$.

Some Properties of periodic functions

1. If T is the period of $f(x)$ then nT is also period of f for any positive integer n.

2. The function $h(x) = af(x) + bg(x)$ has period T if $f(x)$ and $g(x)$ have period T. Here a, b are constants.

3. If $f(x)$ is a periodic function of period T, then $f(ax)$ with $a \neq 0$, is a periodic function of period $\frac{T}{a}$.

4. The period of a sum of a number of periodic functions is the least common multiple of the periods.

5. A constant function is periodic for any period T but has no fundamental period.

Example

Show that $f(x) = \text{const}$ is periodic with any period but has no fundamental period.

Solution

Suppose $f(x) = C$ (const). Now here notice that for any $p > 0$, we have

$$f(x + p) = C = f(x)$$

As, we have chosen p to be arbitrary and hence it shows that $f(x)$ is periodic with any period.
If possible suppose p is the fundamental period of $f(x) = C$ i.e. p is the smallest positive number such that

$$f(x + p) = f(x)$$

but note that $f\left(x + \frac{p}{2}\right) = f(x)$, which contradicts the fact that p is the smallest positive period of f.

Example

Sketch the graph of following functions in $-\pi < x < \pi$

$$f(x) = |\sin x| \qquad \text{and} \qquad g(x) = |x|$$

Solution

We see that the function is defined only one $(-\pi, \pi)$. Thus, the desired graph of $f(x)$ is simply its restriction to the open interval $(-\pi, \pi)$. Graph of $|x|$ is shown in red and graph of $|\sin x|$ is shown in blue.

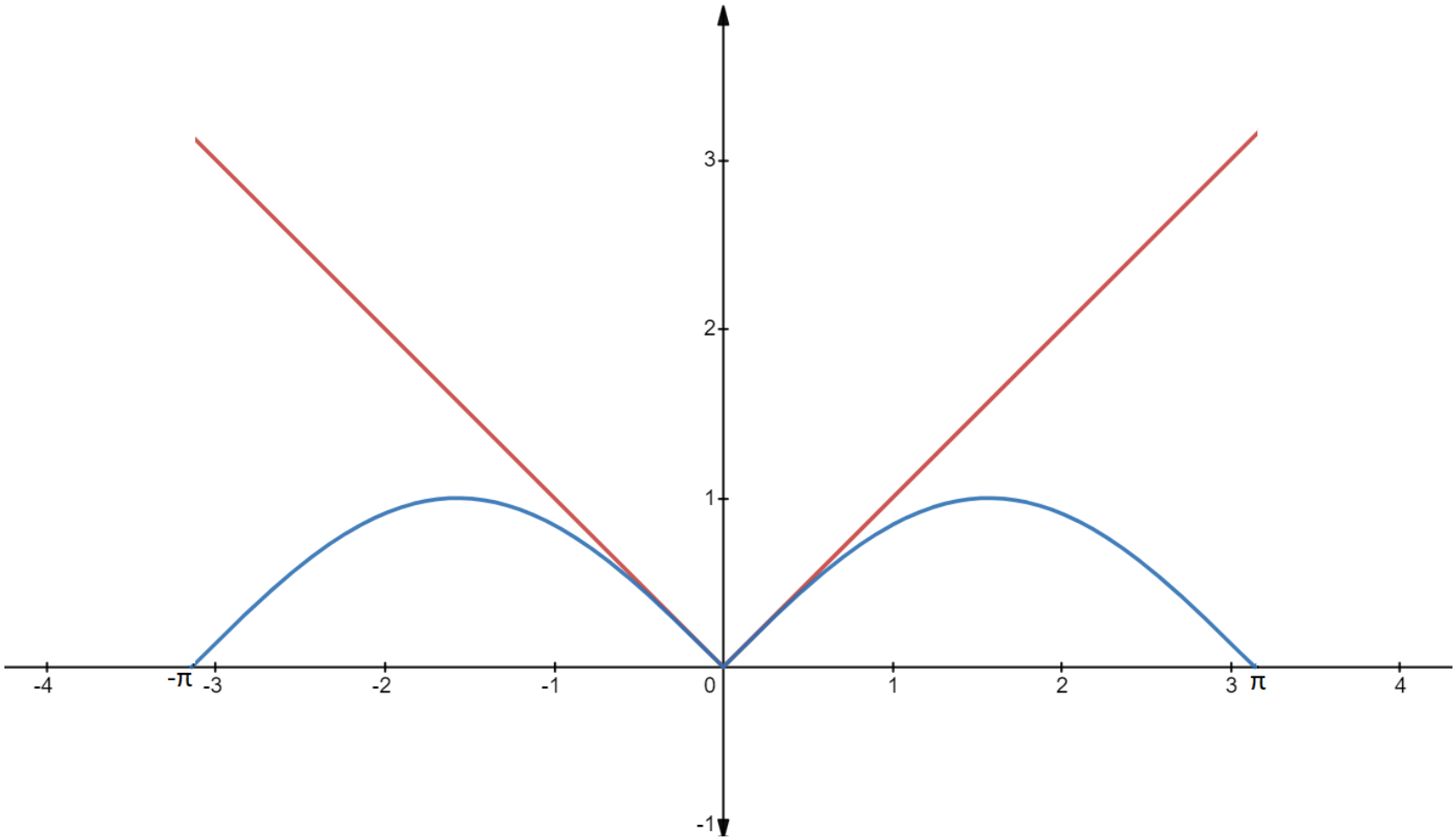


Figure 2: Plot of the functions $f(x) = |\sin x|$ and $g(x) = |x|$

Exercise 1:

The fundamental period is the smallest positive period. Find it for

(a). $\cos x,\ \sin x,\ \cos 2x,\ \sin 2x,\ \cos \pi x,\ \sin \pi x,\ \cos 2\pi x,\ \sin 2\pi x.$

(b). $\cos nx,\ \sin nx,\ \cos \dfrac{2\pi x}{k},\ \sin \dfrac{2\pi x}{k},\ \cos \dfrac{2\pi nx}{k},\ \sin \dfrac{2\pi nx}{k}.$

Exercise 2:

If $f(x)$ and $g(x)$ have period p, show that $h(x) = af(x) + bg(x)$ (a, b, constant) has the period p.

Exercise 3:

Change of scale. If $f(x)$ has period p, show that $f(ax),\ a \neq 0$, and $f(x/b),\ b \neq 0$, are periodic functions of x of periods p/a and bp, respectively. Give examples.

Result

Let m and n be integers such that $m \neq 0, n \neq 0$ then

1. $$\int_{-\pi}^{\pi} \cos nx \cos mx \; dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \end{cases}.$$

2. $$\int_{-\pi}^{\pi} \sin nx \sin mx \; dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \end{cases}.$$

3. $$\int_{-\pi}^{\pi} \cos nx \sin mx \; dx = 0.$$

The above results are valid for or any other interval of length 2π because of periodicity.

Definition(Trigonometric Series)

A trigonometric Series is a functional series of the form

$$\frac{a_0}{2} + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots$$

or

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx),$$

where the constants a_0, $a_n, b_n (n = 1, 2, 3, \ldots)$ are called the coefficients.

Fourier Series

✻ Fourier series are infinite series that represent periodic function in terms of cosines and sines. The series to be obtained will be a trigonometric series, i.e., a series of the form

$$\frac{a_0}{2} + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \qquad (7)$$

where $a_0, a_1, a_2, \ldots, b_1, b_2, \ldots$ are constants, called the coefficients of the series.

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✻ We see that each term is a periodic function of period 2π. Hence, its sum will also be a periodic function of period 2π.

✻ Now suppose that $f(x)$ is a given function of period 2π and is such that it can be represented by series (7). Then using the equality sign, we write

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx) \quad (8)$$

and (8) is called Fourier series of $f(x)$. In this case, coefficients are so called Fourier coefficients of $f(x)$.

* **To determine the coefficient a_0:** Integrate both sides of (8) with respect to x in the interval $-\pi$ to π. Then

$$\begin{aligned}
\int_{-\pi}^{\pi} f(x)dx &= \int_{-\pi}^{\pi} \frac{a_0}{2}\, dx + \int_{-\pi}^{\pi} \left(\sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \right) dx \\
&= \pi a_0 + \sum_{n=1}^{\infty} \left[a_n \int_{-\pi}^{\pi} \cos nx\ dx + b_n \int_{-\pi}^{\pi} \sin nx\ dx \right] \\
&= \pi a_0 \\
\implies a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)dx \qquad (9)
\end{aligned}$$

* **To determine the coefficient a_n:** Multiplying both sides of (8) by $\cos mx$ and integrating with respect to x in the interval $-\pi$ to π, we get

$$\int_{-\pi}^{\pi} f(x)\cos mx\ dx = \int_{-\pi}^{\pi} \frac{a_0}{2}\cos mx dx + \sum_{n=1}^{\infty}\left[a_n \int_{-\pi}^{\pi} \cos nx \cos mx dx + b_n \int_{-\pi}^{\pi} \sin nx \cos mx dx\right]$$

In R.H.S., except the integral of a_n (with $m = n$) all other integrals vanish. For $m = n$, the integral of a_n is π. Thus

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx dx, \quad n = 1, 2, \ldots \tag{10}$$

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* **To determine the coefficient b_n:** Multiplying both sides of (8) by $\sin mx$ and integrating with respect to x in the interval $-\pi$ to π, we get

$$\int_{-\pi}^{\pi} f(x)\sin mx dx = \int_{-\pi}^{\pi} \frac{a_0}{2}\sin mx dx + \sum_{n=1}^{\infty}\left[a_n \int_{-\pi}^{\pi} \cos nx \sin mx dx + b_n \int_{-\pi}^{\pi} \sin nx \sin mx dx\right]$$

In R.H.S., except the integral of b_n (with $m = n$) all other integrals vanish. For $m = n$, the integral of b_n is π. Thus

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx dx, \quad n = 1, 2, \ldots \tag{11}$$

Summary

The Fourier series of a 2π periodic function $f(x)$ in the interval $(-\pi, \pi)$ is given by

Euler's Formula

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

where

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nxdx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nxdx$$

The above expressions of Fourier coefficients are called Euler's formula.

Fourier series of a function $f(x)$ in the interval $(\alpha, \alpha + 2\pi)$

The Fourier series of a 2π periodic function $f(x)$ in the interval $(\alpha, \alpha + 2\pi)$ is given by

Euler's Formula

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

where

$$a_0 = \frac{1}{\pi}\int_{\alpha}^{\alpha+2\pi} f(x)dx$$

$$a_n = \frac{1}{\pi}\int_{\alpha}^{\alpha+2\pi} f(x)\cos nxdx$$

$$b_n = \frac{1}{\pi}\int_{\alpha}^{\alpha+2\pi} f(x)\sin nxdx$$

The above expressions of Fourier coefficients are called Euler's formula.

Dirichlet conditions

Dirichlet conditions for the expansion of $f(x)$ in Fourier series:

Let $f(x)$ be a periodic function with period 2π. Let $f(x)$ be piecewise continuous, and bounded in the interval $(-\pi, \pi)$ with finite number of extrema. Then

✻ At the points of continuity, the Fourier series of $f(x)$ (R.H.S. of (8)) converges to $f(x)$ (L.H.S. of (8)).

✻ At a point of discontinuity x_0 , the Fourier series of $f(x)$ converges to the arithmetic mean of the left and right hand limits of $f(x)$ at x_0 i.e., at

$$\frac{f(x_0 - 0) + f(x_0 + 0)}{2}.$$

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Remark

The Dirichlet conditions are valid for any periodic function defined in any interval.

Example

Find the Fourier coefficients of the following periodic functions

$$f(x) = \begin{cases} -k & \text{if } -\pi < x < 0 \\ k & \text{if } 0 < x < \pi \end{cases} \qquad f(x + 2\pi) = f(x)$$

and hence find the Fourier series of $f(x)$.

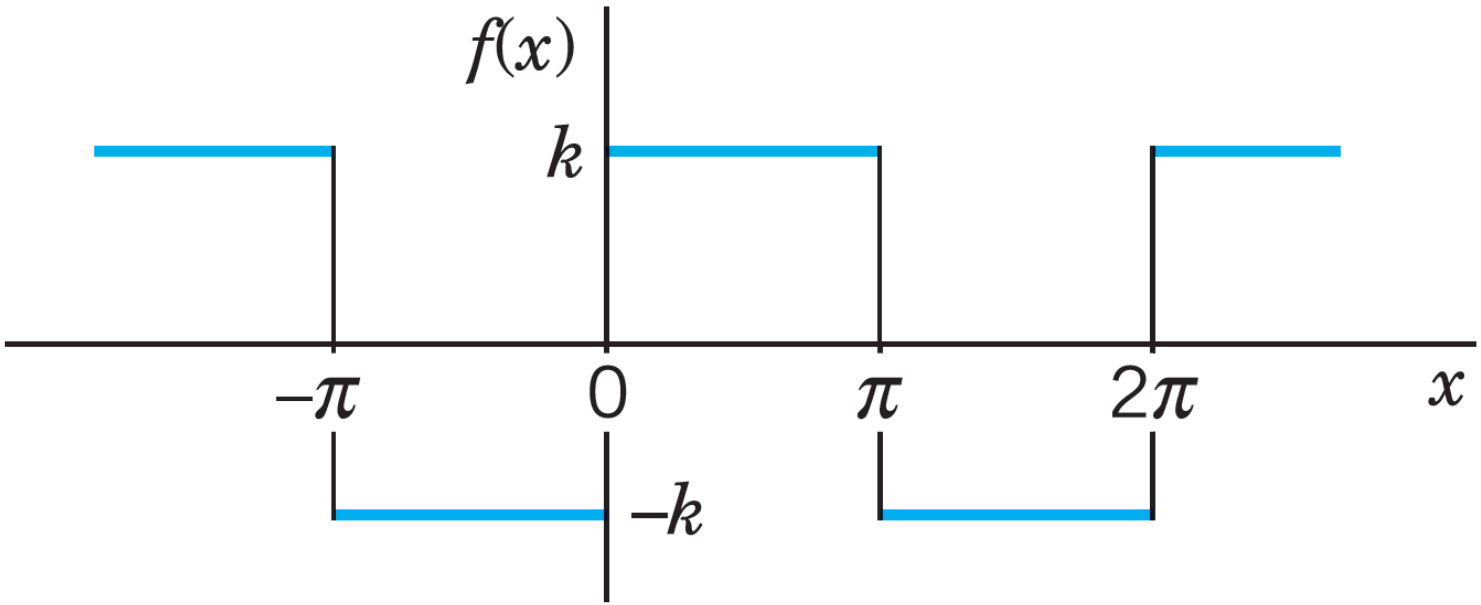


Figure 3: Given function $f(x)$ (Periodic reactangular wave)

Solution

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-k)dx + \int_{0}^{\pi} kdx\right] = -k\pi + k\pi = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nxdx = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-k)\cos nxdx + \int_{0}^{\pi} k\cos nxdx\right] = 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nxdx = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-k)\sin nxdx + \int_{0}^{\pi} k\sin nxdx\right]$$

$$= \frac{1}{\pi}\left[\frac{k\cos nx}{n}\Big|_{-\pi}^{0} - \frac{k\cos nx}{n}\Big|_{0}^{\pi}\right] = \frac{2k}{n\pi}(1-\cos n\pi)$$

So,

$$f(x) = \sum_{n=1}^{\infty}\frac{2k}{n\pi}(1-\cos n\pi)\sin nx$$

$$f(x) = \frac{4k}{\pi}\left[\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \cdots\right] \tag{12}$$

Remark

Putting $x = \frac{\pi}{2}$ in (12) yields

$$f\left(\frac{\pi}{2}\right) = \frac{4k}{\pi}\left[1 - \frac{1}{3} + \frac{1}{5} + \cdots\right]$$

$$\implies k = \frac{4k}{\pi}\left[1 - \frac{1}{3} + \frac{1}{5} + \cdots\right]$$

$$\implies \underbrace{1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7}\cdots}_{\text{an alternating series}} = \frac{\pi}{4}$$

This is a famous result obtained by Leibniz in 1673 from geometric considerations. It illustrates that the values of various series with constant terms can be obtained by evaluating Fourier series at specific points.

Example

Find the fourier series of the following 2π periodic function

$$f(x) = \begin{cases} 1, & 0 < x < \pi, \\ -1, & \pi < x < 2\pi, \end{cases}$$

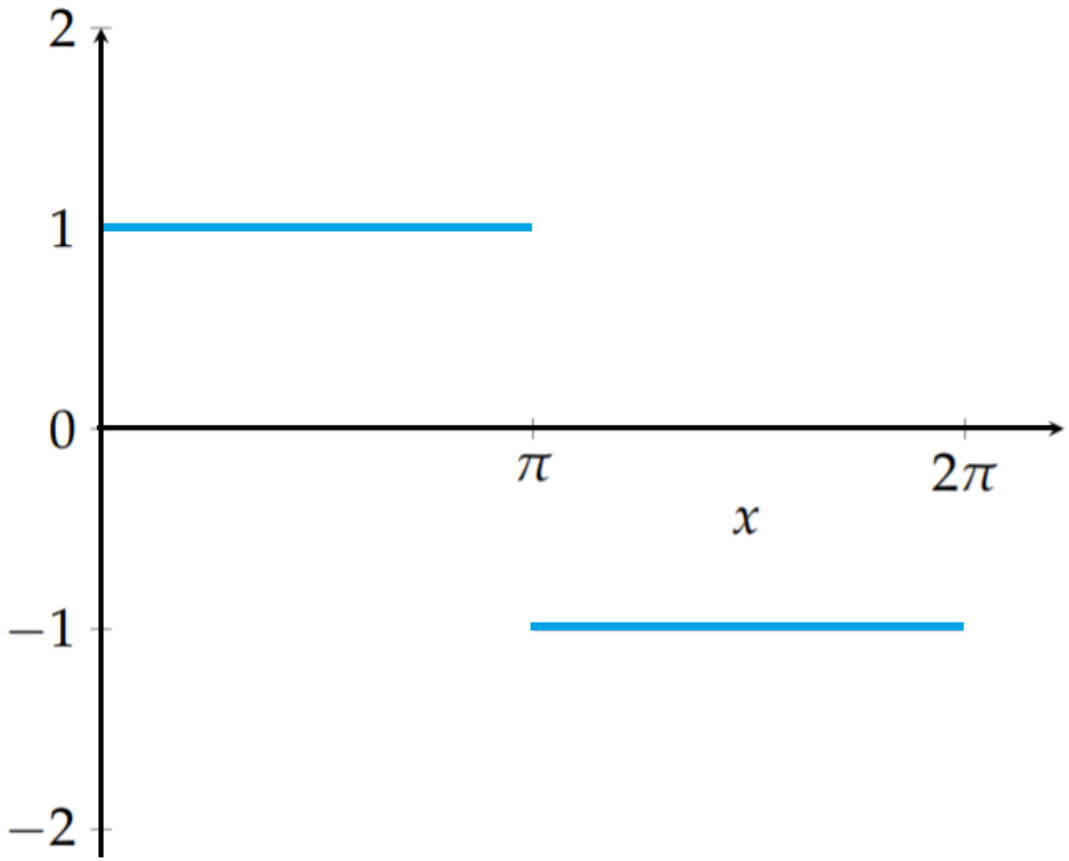


Figure 4: Plot of a discontinuous function

Solution

$$
\begin{aligned}
a_0 &= \frac{1}{\pi}\int_0^{2\pi} f(x)dx \\
&= \frac{1}{\pi}\int_0^{\pi} dx + \frac{1}{\pi}\int_{\pi}^{2\pi}(-1)dx \\
&= \frac{1}{\pi}(\pi) + \frac{1}{\pi}(-2\pi+\pi) = 0.
\end{aligned} \tag{13}
$$

$$
\begin{aligned}
a_n &= \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx dx \\
&= \frac{1}{\pi}\left[\int_0^{\pi}\cos nx dx - \int_{\pi}^{2\pi}\cos nx dx\right] \\
&= \frac{1}{\pi}\left[\left(\frac{1}{n}\sin nx\right)_0^{\pi} - \left(\frac{1}{n}\sin nx\right)_{\pi}^{2\pi}\right] \\
&= 0.
\end{aligned} \tag{14}
$$

Solution cont...

$$
\begin{aligned}
b_n &= \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx dx \\
&= \frac{1}{\pi}\left[\int_0^{\pi} \sin nx dx - \int_{\pi}^{2\pi} \sin nx dx\right] \\
&= \frac{1}{\pi}\left[\left(-\frac{1}{n}\cos nx\right)_0^{\pi} + \left(\frac{1}{n}\cos nx\right)_{\pi}^{2\pi}\right] \\
&= \frac{1}{\pi}\left[-\frac{1}{n}\cos n\pi + \frac{1}{n} + \frac{1}{n} - \frac{1}{n}\cos n\pi\right] \\
&= \frac{2}{n\pi}(1-\cos n\pi). \qquad (15)
\end{aligned}
$$

Therefore, the Fourier series of the given function is

$$
f(x) = \sum_{i=1}^{\infty} \frac{2}{n\pi}(1-\cos n\pi)\sin nx = \frac{4}{\pi}\sum_{\substack{n=1 \\ \text{n odd}}}^{\infty} \frac{1}{n}\sin nx
$$

Example

Find the Fourier series of the following function

$$f(x) = \begin{cases} x, & \text{if} \quad -\frac{\pi}{2} < x < \frac{\pi}{2}, \\ 0 & \text{if} \quad \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}.$$

and hence deduce that

$$\frac{\pi^2}{8} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}.$$

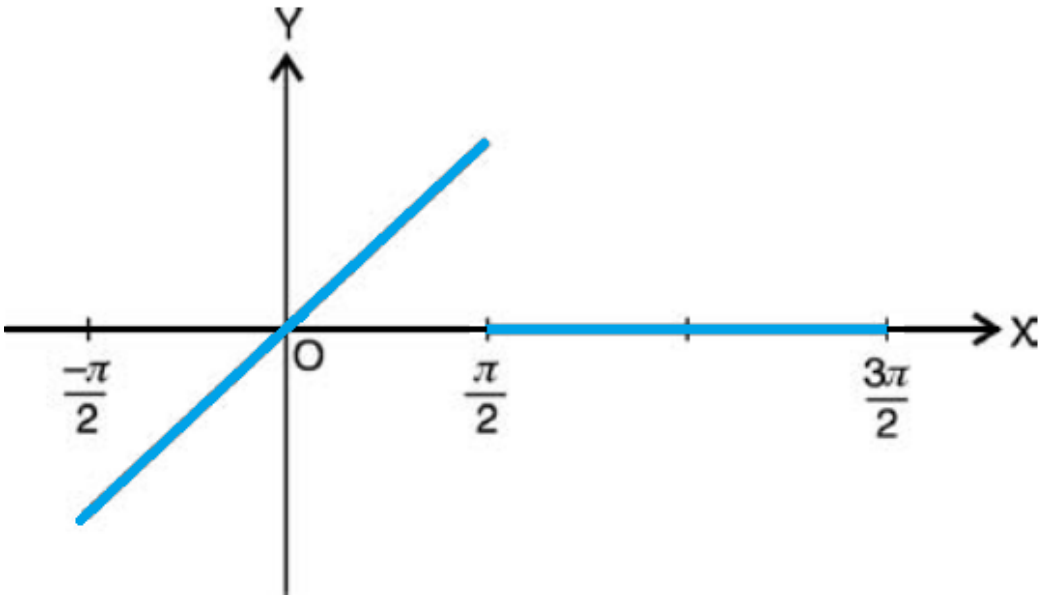


Figure 5: Plot of a discontinuous function

Solution

$$a_0 = \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} f(x)dx = \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} xdx + \frac{1}{\pi}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} 0dx = \frac{1}{\pi}\left[\frac{x^2}{2}\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 0$$

$$a_n = \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} f(x)\cos nxdx = \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x\cos nxdx = \frac{1}{\pi}\left[\frac{x\sin nx}{n} + \frac{\cos nx}{n^2}\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 0$$

$$b_n = \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} f(x)\sin nxdx = \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x\sin nxdx = \frac{1}{\pi}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2\pi}\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= -\frac{1}{n}\cos\left(\frac{n\pi}{2}\right) + \frac{2}{n^2\pi}\sin\left(\frac{n\pi}{2}\right)$$

Solution cont...

Therefore the Fourier series of $f(x)$ is

$$f(x) = \sum_{n=1}^{\infty} \left[-\frac{1}{n} \cos\left(\frac{n\pi}{2}\right) + \frac{2}{n^2\pi} \sin\left(\frac{n\pi}{2}\right) \right] \sin nx.$$

At $x = \frac{\pi}{2}$, which is a point of discontinuity,

$$f\left(\frac{\pi}{2}\right) = \frac{1}{2}\left[f\left(\frac{\pi}{2} - 0\right) + f\left(\frac{\pi}{2} + 0\right)\right] = \frac{1}{2}\left[\frac{\pi}{2} + 0\right] = \frac{\pi}{4}$$

Putting $x = \frac{\pi}{2}$ in the Fourier series expansion yields

$$f\left(\frac{\pi}{2}\right) = \sum_{n=1}^{\infty} \left[-\frac{1}{n} \cos\left(\frac{n\pi}{2}\right) + \frac{2}{n^2\pi} \sin\left(\frac{n\pi}{2}\right) \right] \sin\left(\frac{n\pi}{2}\right)$$

$$\implies \frac{\pi}{4} = \sum_{n=1}^{\infty} \left(0 + \frac{2}{(2n-1)^2\pi}\right)(-1)^{2n}$$

$$\implies \frac{\pi^2}{8} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$$

Example

Obtain the Fourier series of $f(x) = \dfrac{\pi - x}{2}$ in the interval $(0, 2\pi)$ and hence deduce that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots .$$

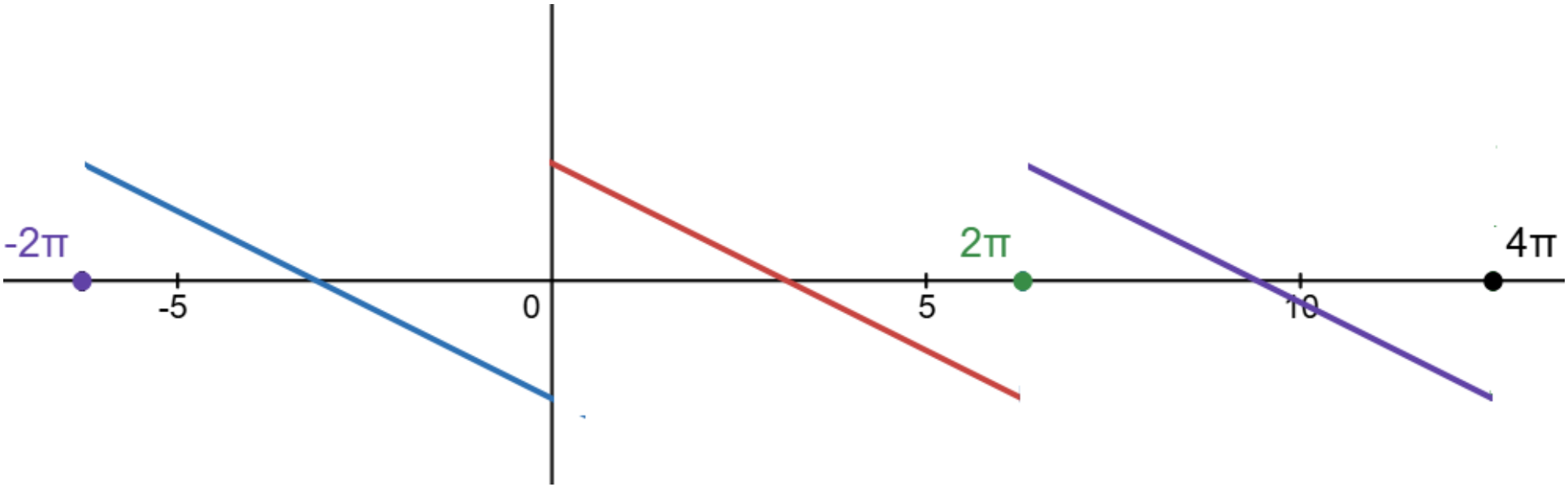


Figure 6: Plot of the periodic function $f(x) = \frac{\pi - x}{2}$

Solution

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)dx = \frac{1}{\pi}\int_0^{2\pi} \frac{\pi - x}{2}dx = \frac{1}{2\pi}\left[\pi x - \frac{x^2}{2}\right]_0^{2\pi} = 0$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx dx = \frac{1}{\pi}\int_0^{2\pi} \left(\frac{\pi - x}{2}\right)\cos nx\ dx$$

$$\implies a_n = \frac{1}{2\pi}\left[(\pi - x)\frac{\sin nx}{n} - \frac{\cos nx}{n^2}\right]_0^{2\pi} = 0$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\ dx = \frac{1}{\pi}\int_0^{2\pi} \frac{\pi - x}{2}\sin nx\ dx$$

$$\implies b_n = \frac{1}{2\pi}\left[(\pi - x)\left(-\frac{cosnx}{n}\right) - \frac{\sin nx}{n^2}\right]_0^{2\pi} = \frac{1}{2\pi}\left[\frac{\pi}{n} + \frac{\pi}{n}\right] = \frac{1}{n}$$

Solution cont...

Therefore, the Fourier series of $f(x)$ is

$$f(x) = \sum_{n=1}^{\infty} \frac{\sin nx}{n}$$

Putting $x = \frac{\pi}{2}$ in above expression yields

$$f\left(\frac{\pi}{2}\right) = \sum_{n=1}^{\infty} \frac{\sin n\left(\frac{\pi}{2}\right)}{n}$$

$$\implies \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

Exercise 4:

Find the Fourier series of the given function $f(x)$, which is assumed to have period 2π. Show the details of your work.

(a). $f(x) = |x|$ on $(-\pi, \pi)$

ans: $f(x) = \frac{\pi}{2} + \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}\cos(2n-1)x$

(b). $f(x) = \begin{cases} 0 & \text{if} \quad -\pi < x < 0 \\ x(\pi - x) & \text{if} \quad 0 < x < \pi \end{cases}$ on $(-\pi, \pi)$

(c). $f(x) = \begin{cases} x & \text{if} \quad -\pi < x < 0 \\ \pi - x & \text{if} \quad 0 < x < \pi \end{cases}$ on $(-\pi, \pi)$

ans: $f(x) = \frac{4}{\pi}\left(\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \cdots\right) + 2\left(\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \cdots\right)$

(d). $f(x) = \begin{cases} -\cos^2 x & \text{if} \quad -\pi < x < 0 \\ \cos^2 x & \text{if} \quad 0 < x < \pi \end{cases}$ on $(-\pi, \pi)$

(e). $f(x) = \begin{cases} x & \text{if} \quad -\pi < x < 0 \\ \pi & \text{if} \quad 0 < x < \pi \end{cases}$ on $(-\pi, \pi)$

ans: $f(x) = \frac{3\pi}{4} - \frac{2}{\pi}\left[\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \cdots\right] - \left[\sin x + \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x + \cdots\right]$

Exercise 5:

Find the Fourier series of the given function $f(x)$, which is assumed to have period 2π. Show the details of your work.

$$f(x) = \begin{cases} \sin 2x & \text{if} \quad -\pi < x < -\dfrac{\pi}{2} \\ 0 & \text{if} \quad -\dfrac{\pi}{2} < x < 0 \\ \sin 2x & \text{if} \quad 0 < x < \pi \end{cases} \qquad \text{on } (-\pi, \pi)$$

Fourier series of functions of arbitrary period

Fourier series of a functions of an arbitrary period:

Let $f(x)$ be a periodic function with period $2L$ defined in the interval $(-L, L)$. Then the Fourier series of $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=0}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right],$$

where the Fourier coefficients of $f(x)$ are given by the Euler formulas

$$a_0 = \frac{1}{L} \int_{-L}^{L} f(x)\, dx$$

$$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos\left(\frac{n\pi x}{L}\right) dx \qquad n = 1, 2, 3, \ldots$$

$$b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right) dx \qquad n = 1, 2, 3, \ldots$$

Example

Find the Fourier series representation of $f(x) = x + 1$ for $-1 \leq x \leq 1$.

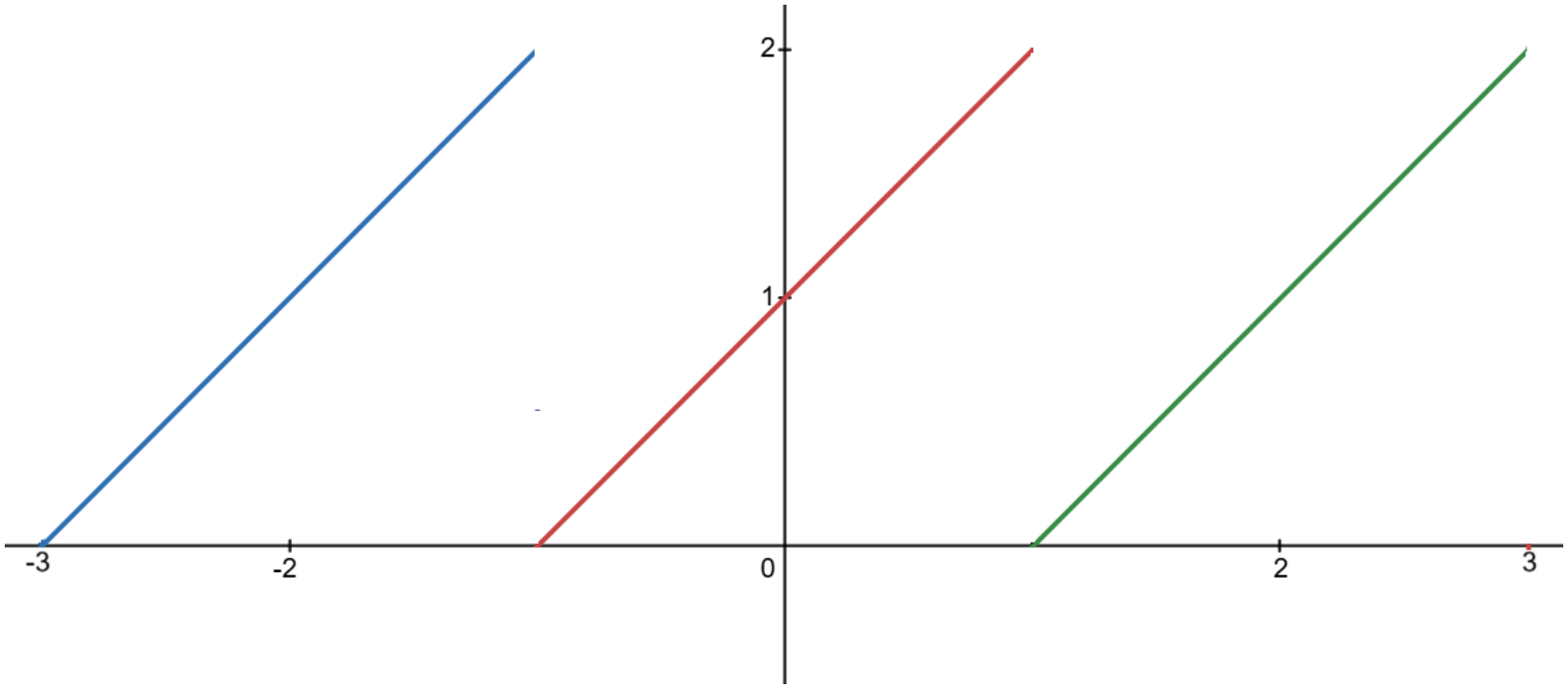


Figure 7: Plot of the function $f(x) = x + 1$

Solution

In this case $2L = 2 \implies L = 1$, so

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\ dx = \int_{-1}^{1}(x+1)\ dx = 2$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx = \int_{-1}^{1}(x+1)\cos n\pi x\ dx$$

$$= \left[\frac{\cos n\pi x}{n^2\pi^2} + \frac{x\sin n\pi x}{n\pi} + \frac{\sin n\pi x}{n\pi}\right]_{-1}^{1} = 0$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\ dx = \int_{-1}^{1}(x+1)\sin n\pi x\ dx$$

$$= \left[\frac{\sin n\pi x}{n^2\pi^2} - \frac{x\cos n\pi x}{n\pi} - \frac{\cos n\pi x}{n\pi}\right]_{-1}^{1} = \frac{2(-1)^{n+1}}{n\pi}$$

Fourier series: $$f(x) = 1 + \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}\sin n\pi x}{n}$$

Example

Find Fourier series expansion of the function

$$f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases} \quad \text{and} \quad f(x + 2\pi) = f(x)$$

And hence deduce that $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$.

Solution

The fourier coefficients of the given function are

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\ dx = \frac{1}{\pi}\int_{-\pi}^{0}(-\pi)\ dx + \frac{1}{\pi}\int_{0}^{\pi} x\ dx = -\pi + \frac{\pi}{2} = -\frac{\pi}{2}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\ dx = \frac{1}{\pi}\int_{-\pi}^{0}(-\pi)\cos nx\ dx + \frac{1}{\pi}\int_{0}^{\pi} x\cos nx\ dx = \frac{(-1)^n - 1}{n^2\pi}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\ dx = \frac{1}{\pi}\int_{-\pi}^{0}(-\pi)\sin nx\ dx + \frac{1}{\pi}\int_{0}^{\pi} x\sin nx\ dx = \frac{1 - 2(-1)^n}{n}$$

And the fourier series is

$$f(x) = -\frac{\pi}{4} + \sum_{n=1}^{\infty}\left(\frac{(-1)^n - 1}{n^2\pi}\right)\cos nx + \sum_{n=1}^{\infty}\left(\frac{1 - 2(-1)^n}{n}\right)\sin nx \qquad (16)$$

Solution cont. . .

Putting $x = 0$ in (16), we have

$$f(0) = -\frac{\pi}{4} + \sum_{n=1}^{\infty}\left(\frac{(-1)^n - 1}{n^2\pi}\right) = -\frac{\pi}{4} - \frac{2}{\pi}\sum_{n=1}^{\infty}\left(\frac{1}{(2n-1)^2}\right) \tag{17}$$

Since $x = 0$ is the point of discontinuity, therefore,

$$f(0) = \frac{L.H.L. + R.H.L.}{2} = \frac{-\pi}{2}.$$

Putting this in (17) and simplifying, we get

$$\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

Exercise 6:

Find the Fourier series of the given function $f(x)$. Show the details of your work.

(a). $f(x) = x + 1$ and $f(x+2) = f(x)$ on $(-1, 1)$

Ans: $f(x) = 1 - \sum_{n=1}^{\infty} \left(\frac{2(-1)^n}{n\pi} \right) \sin n\pi x$

(b). $f(x) = |x|$ and $f(x+2L) = f(x)$ on $(-L, L)$

(c). $f(x) = x$ and $f(x+4) = f(x)$ on $(-2, 2)$

Ans: $f(x) = -\sum_{n=1}^{\infty} \left(\frac{4(-1)^n}{n\pi} \right) \sin \left(\frac{n\pi x}{2} \right)$

(d). $f(x) = 1 - |x|$ and $f(x+2) = f(x)$ on $(-1, 1)$

Ans: $f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \left(\frac{2 - 2(-1)^n}{n^2\pi^2} \cos n\pi x \right)$

Definition (Even and odd functions)

* A function $f(x)$ is said to be even if

$$f(-x) = f(x) \qquad \forall x.$$

* A function $f(x)$ is said to be odd if

$$f(-x) = -f(x) \qquad \forall x.$$

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Example

1. x^2, $\cos x$, $x^4 + \cos 2x + 2$ are examples of even functions.
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Example

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An integration identity

$$\int_{-L}^{L} f(x)\, dx = \begin{cases} 2\int_{0}^{L} f(x)\, dx & \text{if } f(x) \text{ is even,} \\ 0 & \text{if } f(x) \text{ is odd} \end{cases}$$

Fourier Series for Even and Odd Functions

Let the Fourier series of $f(x)$ in $(-L, L)$ be

$$f(x) = \frac{a_0}{2} + \sum_{n=0}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right] \tag{18}$$

where

$$a_0 = \frac{1}{L} \int_{-L}^{L} f(x)\, dx \tag{19}$$

$$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos\left(\frac{n\pi x}{L}\right) dx \tag{20}$$

$$b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right) dx \tag{21}$$

Fourier series of Even function:

Suppose $f(x)$ is even function in $(-L, L)$. Then all b_n's will be zero since the integrand in (21) is an odd function. Thus the Fourier series of an even function contains "only cosine terms" and is known as "**Fourier cosine series**" given by

$$f(x) = \frac{a_0}{2} + \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right),$$

where

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\, dx = \frac{2}{L}\int_{0}^{L} f(x)\, dx$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\, dx = \frac{2}{L}\int_{0}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\, dx$$

Fourier series of Odd function:

Suppose $f(x)$ is odd function in $(-L, L)$. Then all a_n's are zero because the integrand in (20) is an odd function. Also, a_0 is zero since f is odd. Thus the Fourier series of an odd function contains "only sine terms" and is known as "**Fourier sine series**" given by

$$f(x) = \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right),$$

where

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx = \frac{2}{L}\int_{0}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Example

Find the Fourier series of the function $f(x) = |\sin x|$ on $(-\pi, \pi)$, which is assumed to have period 2π. Show the details of your work.

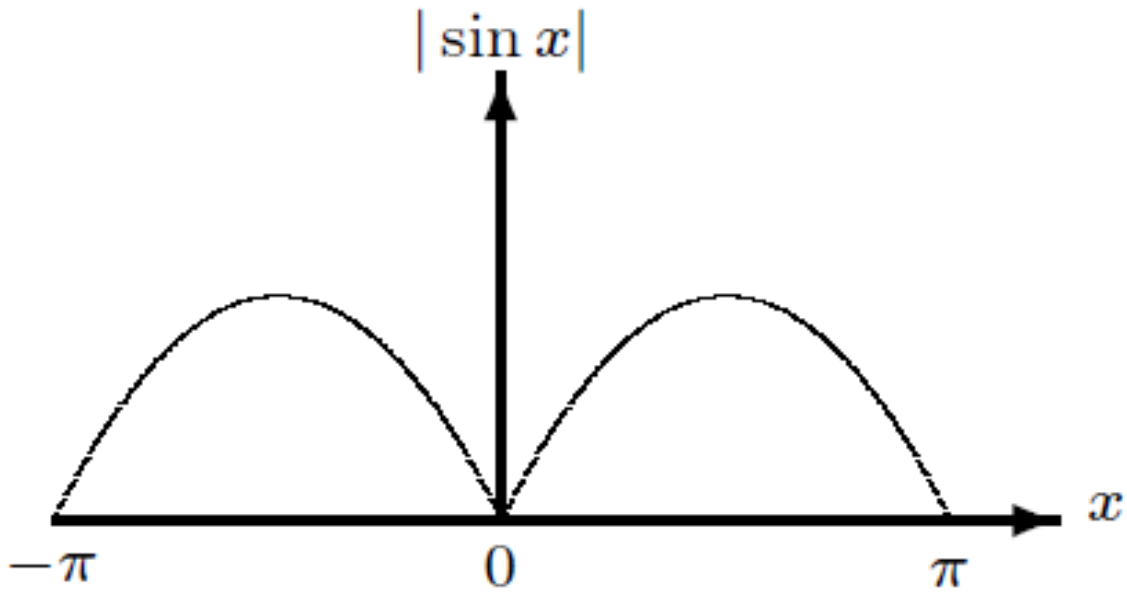


Figure 8: Plot of the function $f(x) = |\sin x|$ on $(-\pi, \pi)$

Solution

Given $f(x) = |\sin x|$ and

$$f(-x) = |\sin(-x)| = |\sin x| = f(x)$$

Hence, $f(x) = |\sin x|$ is an even function. The Fourier series of $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{2}{\pi}\int_0^{\pi} \sin x dx = \frac{2}{\pi}\Big[-cosx\Big]_0^{\pi} = \frac{4}{\pi}$$

and

$$a_n = \frac{2}{\pi}\int_0^{\pi} \sin x \cos nx dx$$

We know that $2\sin x \cos nx = sin(n+1)x + sin(1-n)x$. So, we need to compute a_1 separately.

$$a_1 = \frac{2}{\pi}\int_0^{\pi} \sin x \cos x dx = \frac{1}{\pi}\int_0^{\pi} \sin 2x \ dx = -\frac{\cos 2x}{2\pi}\Big|_0^{\pi} = 0$$

Solution cont...

For $n \geq 2$,

$$a_n = \frac{2}{\pi}\int_0^{\pi} \sin x \cos nx dx = \frac{1}{\pi}\int_0^{\pi} \sin(1+n)x + \sin(1-n)x dx$$

$$= \frac{1}{\pi}\left[\frac{-cos(1+n)x}{1+n} - \frac{cos(1-n)x}{1-n}\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[\frac{-cos(1+n)\pi}{1+n} - \frac{cos(1-n)\pi}{1-n} + \frac{1}{1-n} + \frac{1}{1+n}\right]$$

$$= \frac{1}{\pi}\left[-\frac{(-1)^{n+1}}{n+1} - \frac{(-1)^{n+1}}{1-n} + \frac{1}{1-n} + \frac{1}{1+n}\right]$$

$$= \frac{1-(-1)^{n+1}}{\pi}\left[\frac{1}{1-n} + \frac{1}{1+n}\right] = \frac{2[1-(-1)^{n+1}]}{\pi(1-n^2)}$$

Thus, the Fourier series of $f(x) = |\sin x|$ is

$$f(x) = \frac{2}{\pi} + \sum_{n=2}^{\infty} \frac{2[1-(-1)^{n+1}]}{\pi(1-n^2)} \cos nx.$$

Example

Obtain a Fourier series expression for

$$f(x) = x^3 \qquad \text{for} \qquad -\pi < x < \pi$$

Solution

Since $f(x) = x^3$ is an odd function and hence Fourier series of $f(x)$ is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx dx = \frac{2}{\pi}\int_0^{\pi} x^3 \sin nx dx = \frac{2}{\pi}\left[-x^3\frac{\cos nx}{n}\Big|_0^{\pi} + \int_0^{\pi} 3x^2\frac{\cos nx}{n}dx\right]$$

$$= \frac{2}{\pi}\left[-\frac{(-1)^n\pi^3}{n} + 3x^2\frac{\sin nx}{n^2}\Big|_0^{\pi} - \int_0^{\pi} 6x\frac{\sin nx}{n^2}dx\right]$$

$$= \frac{2}{\pi}\left[-\frac{(-1)^n\pi^3}{n} + 6x\frac{\cos nx}{n^3}\Big|_0^{\pi} - \int_0^{\pi} 6\frac{\cos nx}{n^3}dx\right]$$

$$= \frac{2}{\pi}\left[-\frac{(-1)^n\pi^3}{n} + \frac{6\pi(-1)^n}{n^3} - 6\frac{\sin nx}{n^4}\Big|_0^{\pi}\right] = \frac{2(-1)^n}{n}\left[\frac{6}{n^2} - \pi^2\right]$$

Fourier series: $$f(x) = \sum_{n=1}^{\infty} \frac{2(-1)^n}{n}\left[\frac{6}{n^2} - \pi^2\right]\sin nx$$

Exercise 7:

Are the following functions even or odd or neither even nor odd?

1. e^x, $e^{-|x|}$, $x^3 \cos nx$, $x^2 \tan \pi x$, $\sin x - \cosh x$
2. $\sin^2 x$, $\sin x^2$, $\ln x$, $\dfrac{x}{x^2+1}$, $x \cot x$

Exercise 8:

Prove that the sum of two even functions is an even function and the sum of two odd functions is an odd function.

Exercise 9:

Find all functions that are both even and odd.

Exercise 10:

Prove that if $f(x)$ is an odd function then all the Fourier coefficients $a_n = 0$.

Exercise 11:

Is the given function even or odd or neither even nor odd? Find its Fourier series. Show details of your work.

(a). $f(x) = x^2$ and $f(x+2) = f(x)$ on $(-1, 1)$

Ans: $f(x) = \frac{1}{3} + \frac{4}{\pi^2}\left[-\cos \pi x + \frac{1}{4}\cos 2\pi x - \frac{1}{9}\cos 3\pi x + \cdots\right]$

(b). $f(x) = 1 - \frac{x^2}{4}$ and $f(x+4) = f(x)$ on $(-2, 2)$

Ans: $f(x) = \frac{2}{3} - \sum_{n=1}^{\infty}\left(\frac{4(-1)^n}{n^2\pi^2}\right)\cos\left(\frac{n\pi x}{2}\right)$

(c). $f(x) = \cos \pi x$ and $f(x+1) = f(x)$ on $\left(-\frac{1}{2}, \frac{1}{2}\right)$

Ans: $f(x) = \frac{2}{\pi} + 2\sum_{n=1}^{\infty}\left[\frac{\sin\left(\frac{(2n+1)\pi x}{2}\right)}{(2n+1)\pi} + \frac{\sin\left(\frac{(2n-1)\pi x}{2}\right)}{(2n-1)\pi}\right]\cos 2n\pi x$

Exercise 12:

Is the given function even or odd or neither even nor odd? Find its Fourier series. Show details of your work.

(a). $f(x) = x|x|$ and $f(x+2) = f(x)$ on $(-1, 1)$

Ans: $f(x) = \sum_{n=1}^{\infty} \left(\frac{-4 + 4(-1)^n}{n^3\pi^3} - \frac{2(-1)^n}{n\pi} \right) \sin n\pi x$

(b). $f(x) = |x|$ and $f(x+2L) = f(x)$ on $(-L, L)$

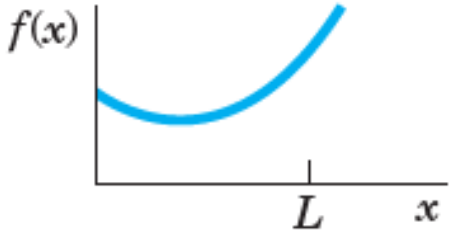
Half-Range Expansions

❀ So far we have considered the Fourier series expansion of a function which is periodic, defined in an interval $-L$ to L of length $2L$. Now, we consider the procedure to expand a non-periodic function $f(x)$ defined in half of the above interval say $(0, L)$ of length L. Such expansions are known as half range expansions or half range Fourier series.

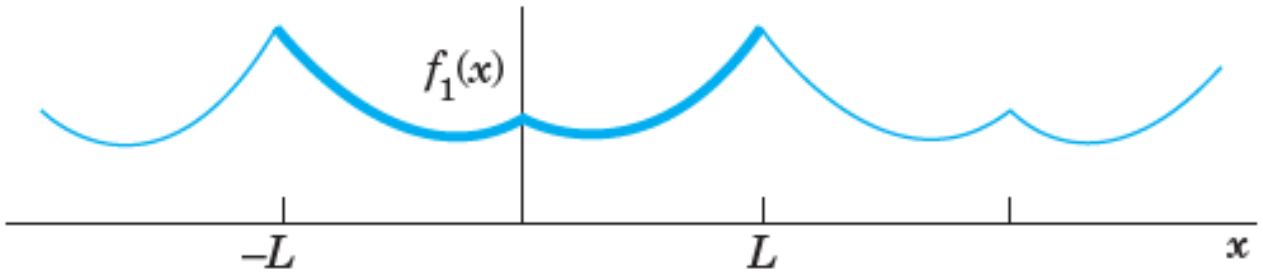
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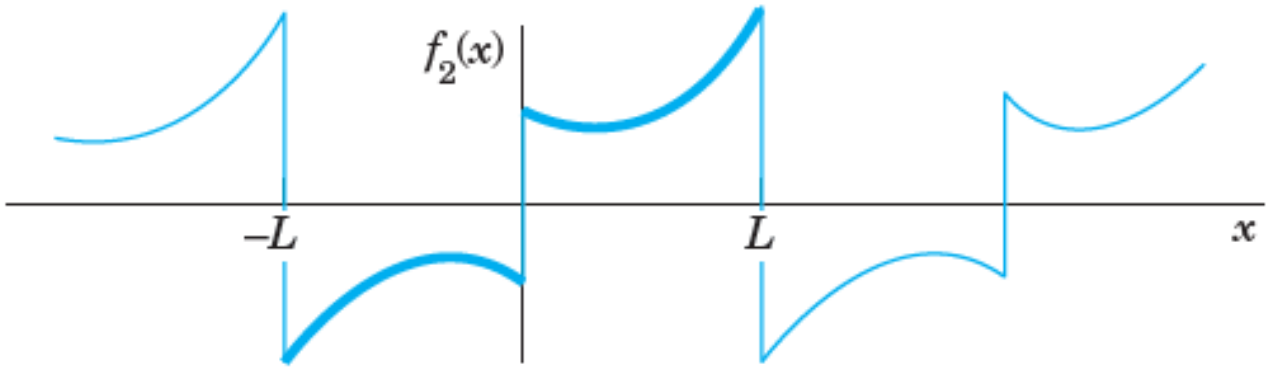
❀ The given function is defined in the interval $(0, L)$ and it is immaterial whatever the function may be outside the interval $(0, L)$. We can extend $f(x)$ as even function or as an odd function in the interval $(-L, L)$.



(0) The given function $f(x)$



(a) $f(x)$ continued as an ***even*** periodic function of period $2L$



(b) $f(x)$ continued as an ***odd*** periodic function of period $2L$

Figure 9: Even and odd extensions of period $2L$

Half-Range Expansions cont. . .

❀ To get the series of cosines only, we assume that $f(x)$ is an even function in the interval $(-L, L)$.

$$f(x) = \frac{a_0}{2} + \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right),$$

where

$$a_0 = \frac{2}{L}\int_0^L f(x)\, dx$$

and

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\, dx$$

Half-Range Expansions cont. . .

❀ To expand $f(x)$ as a sine series, we extend the function in the interval $(-L, L)$ as an odd function.

$$f(x) = \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right),$$

where

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\, dx$$

Example

Obtain Half range cosine series expansion of the function

$$f(x) = x^2, \quad 0 < x < 2$$

And hence deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Solution

Half range fourier cosine series of $f(x) = x^2, \quad 0 < x < 2$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

where

$$a_0 = \frac{2}{L}\int_0^L f(x)dx = \frac{2}{2}\int_0^2 x^2 dx = \frac{8}{3}$$

and

$$\begin{aligned} a_n &= \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right) \\ &= \frac{1}{2}\int_0^2 x^2 \cos\left(\frac{n\pi x}{2}\right) dx \\ &= \frac{16(-1)^n}{n^2\pi^2} \end{aligned}$$

Solution cont. . .

Thus fourier cosine series of x^2 is

$$x^2 = \frac{4}{3} + \sum_{n=1}^{\infty} \frac{16(-1)^n}{n^2\pi^2} \cos\left(\frac{n\pi x}{2}\right)$$

Putting $x = 2$ in above expression, we have

$$4 = \frac{4}{3} + \sum_{n=1}^{\infty} \frac{16(-1)^n}{n^2\pi^2} \cos n\pi$$

$$\implies \frac{8}{3} = \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\implies \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Example

Find the Fourier sine series for the function

$$f(x) = e^{ax} \qquad \text{for} \qquad 0 < x < \pi,$$

where a is a constant.

Solution

$$
\begin{aligned}
b_n &= \int_0^{\pi} e^{ax} \sin nx dx \\
&= \frac{2}{\pi}\left[\frac{e^{ax}}{a^2+n^2}(a\sin nx - n\cos nx)\right]_0^{\pi} \\
&= \frac{2}{\pi}\left[\frac{e^{a\pi}}{a^2+n^2}(-n\cos n\pi) + \frac{n}{a^2+n^2}\right] \\
&= \frac{2n}{a^2+n^2}[1-(-1)^n e^{a\pi}]
\end{aligned}
$$

Therefore the Fourier sine series of the given function is

$$
f(x) = \sum_{n=1}^{\infty} \frac{2n}{a^2+n^2}[1-(-1)^n e^{a\pi}]\sin nx.
$$

Example

Find the Fourier cosine and sine series expansion of the following function

$$f(x) = \begin{cases} \frac{2k}{L}x & \text{if } 0 < x < \frac{L}{2} \\ \\ \frac{2k}{L}(L-x) & \text{if } \frac{L}{2} < x < L. \end{cases}$$

Solution

$$a_0 = \frac{2}{L}\int_0^L f(x)dx = \frac{2}{L}\left[\int_0^{\frac{L}{2}} \frac{2k}{L}x\ dx + \int_{\frac{L}{2}}^{L} \frac{2k}{L}(L-x)dx\right]$$

$$= \frac{4k}{L^2}\left[\int_0^{\frac{L}{2}} x\ dx + \int_{\frac{L}{2}}^{L}(L-x)\ dx\right] = \frac{4k}{L^2}\left[\frac{x^2}{2}\Big|_0^{\frac{L}{2}} + \left(Lx - \frac{x^2}{2}\right)\Big|_{\frac{L}{2}}^{L}\right] = k$$

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)dx = \frac{4k}{L^2}\left[\int_0^{\frac{L}{2}} x\cos\left(\frac{n\pi x}{L}\right)dx + \int_{\frac{L}{2}}^{L}(L-x)\cos\left(\frac{n\pi x}{L}\right)dx\right]$$

$$= \frac{4k}{n^2\pi^2}\left(2\cos\frac{n\pi}{2} - \cos n\pi - 1\right)$$

Thus the Fourier cosine series of the given function is

$$f(x) = \frac{k}{2} + \sum_{n=1}^{\infty}\frac{4k}{n^2\pi^2}\left(2\cos\frac{n\pi}{2} - \cos n\pi - 1\right)\cos\left(\frac{n\pi x}{L}\right)$$

Solution cont...

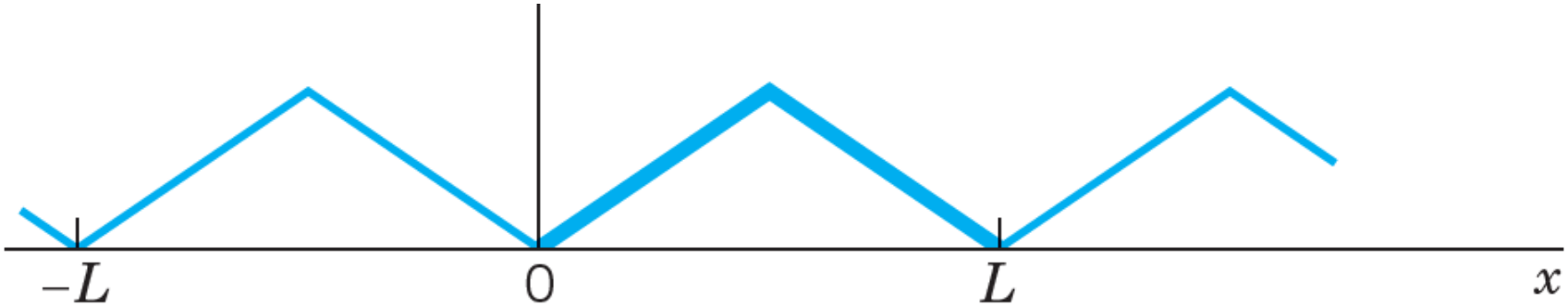
$$b_n = \frac{2}{L}\int_0^L f(x)\,dx = \frac{4k}{L^2}\left[\int_0^{\frac{L}{2}} x\sin\left(\frac{n\pi x}{L}\right)dx + \int_{\frac{L}{2}}^{L}(L-x)\sin\left(\frac{n\pi x}{L}\right)dx\right]$$

$$= \frac{8k}{n^2\pi^2}\sin\frac{n\pi}{2}$$

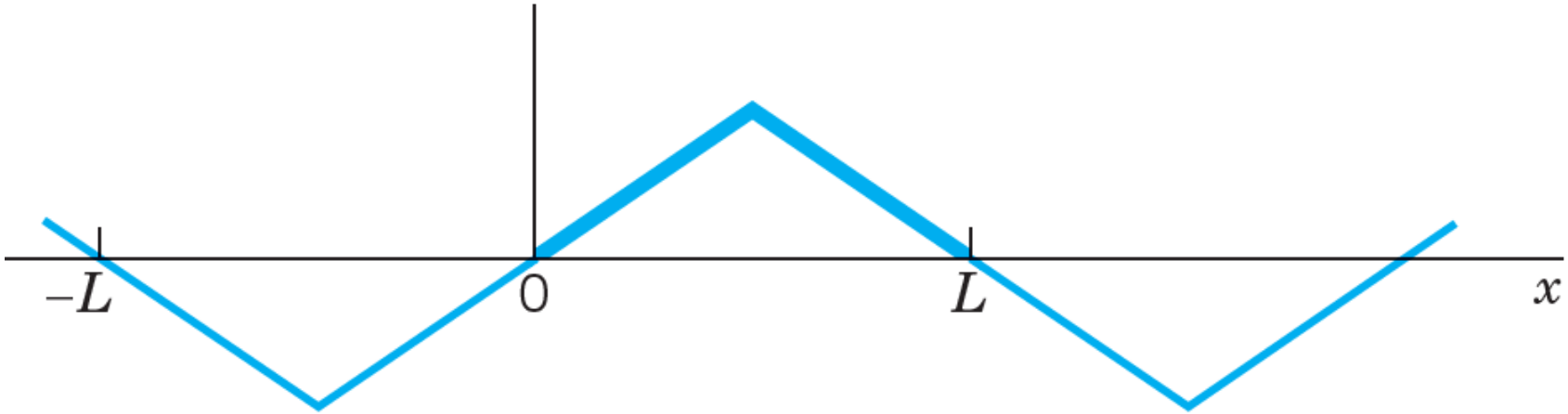
Thus the Fourier cosine series of the given function is

$$f(x) = \sum_{n=1}^{\infty}\frac{8k}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)\sin\left(\frac{n\pi x}{L}\right)$$

Solution cont. . .



(a) Even Extension/ Cosine Extension



(b) Odd Extension/ Sine Extension

Figure 10: Even and odd periodic extensions of $f(x)$

Exercise 13:

Find the half range Fourier cosine and sine series expansion of following functions

(a). $f(x) = x \qquad 0 < x < \pi$

Ans:

$$\text{half range Fourier cosine Series} = \frac{\pi}{2} + \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\cos(2n-1)x}{(2n-1)^2}$$

$$\text{half range Fourier cosine Series} = \sum_{n=1}^{\infty}\frac{2(-1)^{n+1}}{n}\sin nx$$

(b). $f(x) = \sin x \qquad 0 < x < \pi$

Ans:

$$\text{half range Fourier cosine Series} = \frac{2}{\pi} - \frac{4}{\pi}\left(\frac{\cos 2x}{3} + \frac{\cos 4x}{15} + \frac{\cos 6x}{35} + \cdots\right)$$

$$\text{half range Fourier cosine Series} = \sin x$$

Exercise 14:

Find the half range Fourier cosine and sine series expansion of following functions

(a). $f(x) = e^{-x} \qquad 0 < x < \pi$

Ans:

$$\text{half range Fourier cosine Series} = \frac{1-e^{-\pi}}{\pi} - \frac{2}{\pi}\sum_{n=1}^{\infty}\left(\frac{1-(-1)^n e^{-\pi}}{n^2+1}\right)\cos nx$$

$$\text{half range Fourier cosine Series} = \frac{2}{\pi}\sum_{n=1}^{\infty} n\left(\frac{1-(-1)^n e^{-\pi}}{n^2+1}\right)\sin nx$$

(b). $f(x) = e^{x} \qquad 0 < x < \pi$

Ans:

$$\text{half range Fourier cosine Series} = \frac{e^{\pi}-1}{\pi} - \frac{2}{\pi}\sum_{n=1}^{\infty}\left(\frac{1-(-1)^n e^{\pi}}{n^2+1}\right)\cos nx$$

$$\text{half range Fourier cosine Series} = \frac{2}{\pi}\sum_{n=1}^{\infty} n\left(\frac{1-(-1)^n e^{\pi}}{n^2+1}\right)\sin nx$$

Parseval's identity

Suppose $f(x)$ is a square integrable function over $[-L, L]$, i.e., $f(x)$ and $f^2(x)$ are integrable over $[-L, L]$ with the Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{n=0}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right],$$

Then

$$\frac{{a_0}^2}{2} + \sum_{n=1}^{\infty} {a_n}^2 + {b_n}^2 = \frac{1}{L} \int_{-L}^{L} f^2(x)\, dx \tag{22}$$

The (22) is called **Parseval's identity**.

Example

Find the Fourier series for $f(x) = x^2$ in $-\pi < x < \pi$ and hence deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

Solution

Since $f(x) = x^2$ is an even function and hence the Fourier series of $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right),$$

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\, dx = \frac{2}{\pi}\int_0^{\pi} x^2\, dx = \frac{2\pi^2}{3}$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx)\, dx = \frac{2}{\pi}\int_0^{\pi} x^2\cos(nx)\, dx = \frac{4}{n^2}(-1)^n$$

Therefore the Fourier series of $f(x)$ is

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2}(-1)^n \cos nx$$

Solution cont. . .

Now by Parseval's identity, we have

$$\frac{{a_0}^2}{2} + \sum_{n=1}^{\infty} {a_n}^2 + {b_n}^2 = \frac{1}{L}\int_{-L}^{L} f^2(x)\, dx$$

$$\implies \frac{1}{2}\left(\frac{2\pi^2}{3}\right)^2 + \sum_{n=1}^{\infty}\left(\frac{4}{n^2}(-1)^n\right)^2 = \frac{1}{\pi}\int_{-\pi}^{\pi} x^4 dx$$

$$\implies \frac{2\pi^4}{9} + 16\sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{2\pi^4}{5}$$

$$\implies \sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{90}$$

Thus

$$\boxed{\sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{90}}$$

Example

Let $f(x) = |x|$ for x in $[-\pi, \pi]$.

(a) Computer the Fourier Series of $f(x)$ on $[-\pi, \pi]$.

(b) Use the result of part (a) to show that

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$$

Solution

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} |x|dx = \frac{2}{\pi}\int_{0}^{\pi} xdx = \pi$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} |x|\cos nx\, dx = \frac{2}{\pi}\int_{0}^{\pi} x\cos nx\, dx$$

$$= \overbrace{\frac{2}{n\pi}x\sin nx\Big|_0^{\pi}}^{=0} - \frac{2}{n\pi}\int_0^{\pi}\sin nx\, dx$$

$$= \frac{2}{n^2\pi}\left((-1)^n - 1\right) = \begin{cases} 0 & , \ n \text{ is even} \\ -\frac{4}{n^2\pi} & , \ n \text{ is odd} \end{cases}$$

$$f(x) = |x| = \frac{\pi}{2} - \sum_{n=1}^{\infty}\frac{4}{(2n-1)^2\pi}\cos nx$$

$$\frac{{a_0}^2}{2} + \sum_{n=1}^{\infty} {a_n}^2 + {b_n}^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x)\ dx$$

$$\implies \frac{\pi^2}{2} + \sum_{n=1}^{\infty} \left[\frac{2}{n^2\pi} \left((-1)^n - 1 \right) \right]^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2\ dx$$

$$\implies \frac{1}{2}\pi^2 + \sum_{n=1}^{\infty} \frac{16}{(2n-1)^4\pi^2} = \frac{2\pi^2}{3}$$

$$\implies \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^2}{6}$$

$$\implies \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$$

Thus

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}}$$

Example

Find the half range Fourier sine series expansion of $f(x) = x$ in [0,2] and hence deduce that

$$\sum_{m=0}^{\infty} \frac{1}{(2m+1)^2} = \frac{\pi^2}{8}.$$

Solution

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx = \frac{2}{2}\int_0^2 f(x)\sin\left(\frac{n\pi x}{2}\right)\,dx = \frac{4(-1)^{n+1}}{n\pi}$$

Therefore half range Fourier sine series expansion of $f(x)$

$$f(x) = x = \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\left(\frac{n\pi x}{2}\right).$$

By Parseval's identity, we have

$$\frac{{a_0}^2}{2} + \sum_{n=1}^{\infty} {a_n}^2 + {b_n}^2 = \frac{1}{L}\int_{-L}^{L} f^2(x)\,dx$$

$$\implies \sum_{n=1}^{\infty}\left[\frac{4(-1)^{n+1}}{n\pi}\right]^2 = \frac{1}{2}\int_{-2}^{2} x^2\,dx = \frac{2}{2}\int_0^2 x^2\,dx = \frac{8}{3}$$

$$\implies \sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}$$

Solution cont. . .

Also, note that

$$\sum_{n=1}^{\infty}\frac{1}{n^2}=\overbrace{\sum_{m=1}^{\infty}\frac{1}{(2m)^2}}^{\text{Even terms}}+\overbrace{\sum_{m=0}^{\infty}\frac{1}{(2m+1)^2}}^{\text{Odd terms}}$$

$$\implies \frac{\pi^2}{6}=\frac{1}{4}\sum_{m=1}^{\infty}\frac{1}{m^2}+\sum_{m=0}^{\infty}\frac{1}{(2m+1)^2}$$

$$\implies \frac{\pi^2}{6}=\frac{1}{24}\frac{\pi^2}{6}+\sum_{m=0}^{\infty}\frac{1}{(2m+1)^2}$$

$$\implies \sum_{m=0}^{\infty}\frac{1}{(2m+1)^2}=\frac{\pi^2}{6}-\frac{\pi^2}{24}=\frac{\pi^2}{8}$$

$$\sum_{m=0}^{\infty}\frac{1}{(2m+1)^2}=\frac{\pi^2}{8}.$$

Exercise 15:

The function $f(x) = x^3 - \pi^2 x$ has the Fourier series

$$f(x) = \sum_{n=1}^{\infty} \frac{12(-1)^n}{n^3} \sin nx.$$

What is the value of $\sum_{n=1}^{\infty} \frac{1}{n^6}$?

Ans: $\sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}$

Exercise 16:

State and prove Parseval's identity.

Exercise 17:

Given the Fourier series $t^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nt$, $-\pi < t < \pi$, deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

Computation of harmonics

Sometimes the function is not given by a formula, but by a graph or by a table of corresponding values. The process of finding the Fourier series for a function given by such values of the function and independent variable is known as Harmonic Analysis. The Fourier constants are evaluated by the following Euler's formulae:

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = 2\frac{1}{\pi-(-\pi)}\int_{-\pi}^{\pi} f(x)dx = 2[\text{mean of } f(x) \text{ in } (-\pi,\pi)]$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx dx = 2\frac{1}{\pi-(-\pi)}\int_{-\pi}^{\pi} f(x)\cos nx dx = 2[\text{mean of } f(x)\cos nx \text{ in } (-\pi,\pi)]$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx dx = 2\frac{1}{\pi-(-\pi)}\int_{-\pi}^{\pi} f(x)\sin nx dx = 2[\text{mean of } f(x)\sin nx \text{ in } (-\pi,\pi)]$$

Fundamental or first harmonic: The term $a_1\cos x + b_1\sin x$ in Fourier series is called the fundamental or first harmonic.

Second harmonic: The term $a_2\cos 2x + b_2\sin 2x$ in Fourier series is called the second harmonic and so on.

Example

Find the Fourier series as far as the second harmonic to represent the function given by table below:

x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°
f(x)	2.34	3.01	3.69	4.15	3.69	2.20	0.83	0.51	0.88	1.09	1.19	1.64

Solution

x°	$\sin x$	$\sin 2x$	$\cos x$	$\cos 2x$	f(x)	$f(x)\cdot \sin x$	$f(x)\cdot \sin 2x$	$f(x)\cdot \cos x$	$f(x)\cdot \cos 2x$
0	0	0	1	1	2.34	0	0	2.340	2.340
30	0.50	0.87	0.87	0.50	3.01	1.505	2.619	2.619	1.505
60	0.87	0.87	0.50	-0.50	3.69	3.210	3.210	1.845	-1.845
90	1.00	0	0	-1.00	4.15	4.150	0	0	-4.150
120	0.87	-0.87	-0.50	-0.50	3.69	3.210	-3.210	-1.845	-1.845
150	0.50	-0.87	-0.87	0.50	2.20	1.100	-1.914	-1.914	1.100
180	0	0	-1	1.00	0.83	0	0	-0.830	0.830
210	-0.50	0.87	-0.87	0.50	0.51	-0.255	0.444	-0.444	0.255
240	-0.87	0.87	-0.50	-0.50	0.88	-0.766	0.766	-0.440	-0.440
270	-1.00	0	0	-1.00	1.09	-1.090	0	0	-1.090
300	-0.87	-0.87	0.50	-0.50	1.19	-1.035	-1.035	0.595	-0.595
330	-0.50	-0.87	0.87	0.50	1.64	-0.820	- 1.427	1.427	0.820
					25.22	9.209	-0.547	3.353	-3.115

Solution cont...

$$a_0 = 2 \times \text{Mean of } f(x) = 2 \times \frac{25.22}{12} = 4.203$$

$$a_1 = 2 \times \text{Mean of } f(x)\cos x = 2 \times \frac{3.353}{12} = 0.559$$

$$a_2 = 2 \times \text{Mean of } f(x)\cos 2x = 2 \times \frac{-3.115}{12} = -0.519$$

$$b_1 = 2 \times \text{Mean of } f(x)\sin x = 2 \times \frac{9.209}{12} = 1.535$$

$$a_0 = 2 \times \text{Mean of } f(x)\sin 2x = 2 \times \frac{-0.547}{12} = -0.091$$

Fourier series is

$$f(x) = \frac{a_0}{2} + a_1\cos x + a_2\cos 2x + \cdots + b_1\sin x + b_2\sin 2x + \cdots$$
$$= 2.1015 + 0.559\cos x - 0.519\cos 2x + \cdots + 1.535\sin x - 0.091\sin 2x + \cdots$$

Exercise 18:

A machine completes its cycle of operations every time as certain pulley completes a revolution. The displacement $f(x)$ of a point on a certain portion of the machine is given in the table given below for twelve positions of the pulley, x being the angle in degree turned through by the pulley. Find a Fourier series to represent $f(x)$ for all values of x.

x	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
f(x)	7.976	8.026	7.204	5.676	3.674	1.764	0.552	0.262	0.904	2.492	4.736	6.824

Ans:

$f(x) = 4.17 + 2.45\cos x + 0.12\cos 2x + 0.08\cos 3x + \cdots + 3.16\sin x + 0.03\sin 2x + 0.01\sin 3x + \cdots$

Frequency and amplitude spectra of a function

We have discussed how for a periodic function $f(t)$ with period $T = 2L$ and fundamental frequency $f_0 = \frac{1}{T}$, the Fourier series is a representation of the function in terms of sine and cosine functions as follows:

$$f(t) = \frac{a_0}{2} + \sum_{n=0}^{\infty} \left[a_n \cos(2n\pi f_0 t) + b_n \sin(2n\pi f_0 t) \right], \tag{23}$$

where a_n and b_n are coefficients defined as integrals in terms of the specific $f(t)$.

❃ Note that a cosine term $a_n \cos(2n\pi f_0 t)$ and $b_n \sin(2n\pi f_0 t)$ (of the same frequency nf_0) may be viewed as a single cosine waveform of frequency nf_0 :

$$\begin{aligned}
&a_n \cos(2n\pi f_0 t) + b_n \sin(2n\pi f_0 t) \\
=&\sqrt{{a_n}^2 + {b_n}^2} \left[\frac{a_n}{\sqrt{{a_n}^2 + {b_n}^2}} \cos(2n\pi f_0 t) + \frac{b_n}{\sqrt{{a_n}^2 + {b_n}^2}} \sin(2n\pi f_0 t) \right] \\
=&\sqrt{{a_n}^2 + {b_n}^2} \left[\cos\phi_n \cos(2n\pi f_0 t) - \sin\phi_n \sin(2n\pi f_0 t) \right] = A_n \cos(2n\pi f_0 t + \phi_n),
\end{aligned}$$

where the amplitude $A_n = \sqrt{{a_n}^2 + {b_n}^2}$ and the phase angle $\phi_n == -\tan^{-1}\left(\frac{b_n}{a_n}\right)$.

Cont. . .

Thus, we may alternatively write (23) as

$$f(t) = \frac{a_0}{2} + \sum_{n=0}^{\infty} A_n \cos\left(2n\pi f_0 t + \phi_n\right), \tag{24}$$

where $A_n = \sqrt{{a_n}^2 + {b_n}^2}$ and $\phi_n == -\tan^{-1}\left(\frac{b_n}{a_n}\right)$.

Cont. . .

✾ In any Fourier series for a real periodic function, each pair of a_n and b_n coefficients leads to a single cosine with frequency nf_0. The phase ϕ_n of each cosine may be different, just as the non-negative amplitudes A_n are generally different. The phase relationships are important because they correspond to having different amounts of "time shifts" or "delays" for each of the sinusoidal waveforms relative to a zero-phase waveform.
Illustrating the importance of **phase**, in the figure below are shown two waveforms,

$$\begin{aligned} X_1(t) &= \cos(2\pi 2 f_0 t) - 0.5\cos\left(2\pi 3 f_0 t + \frac{\pi}{4}\right) \\ &= \cos(2\pi 2 f_0 t) + 0.5\cos\left(2\pi 3 f_0 t + \frac{5\pi}{4}\right) \end{aligned}$$

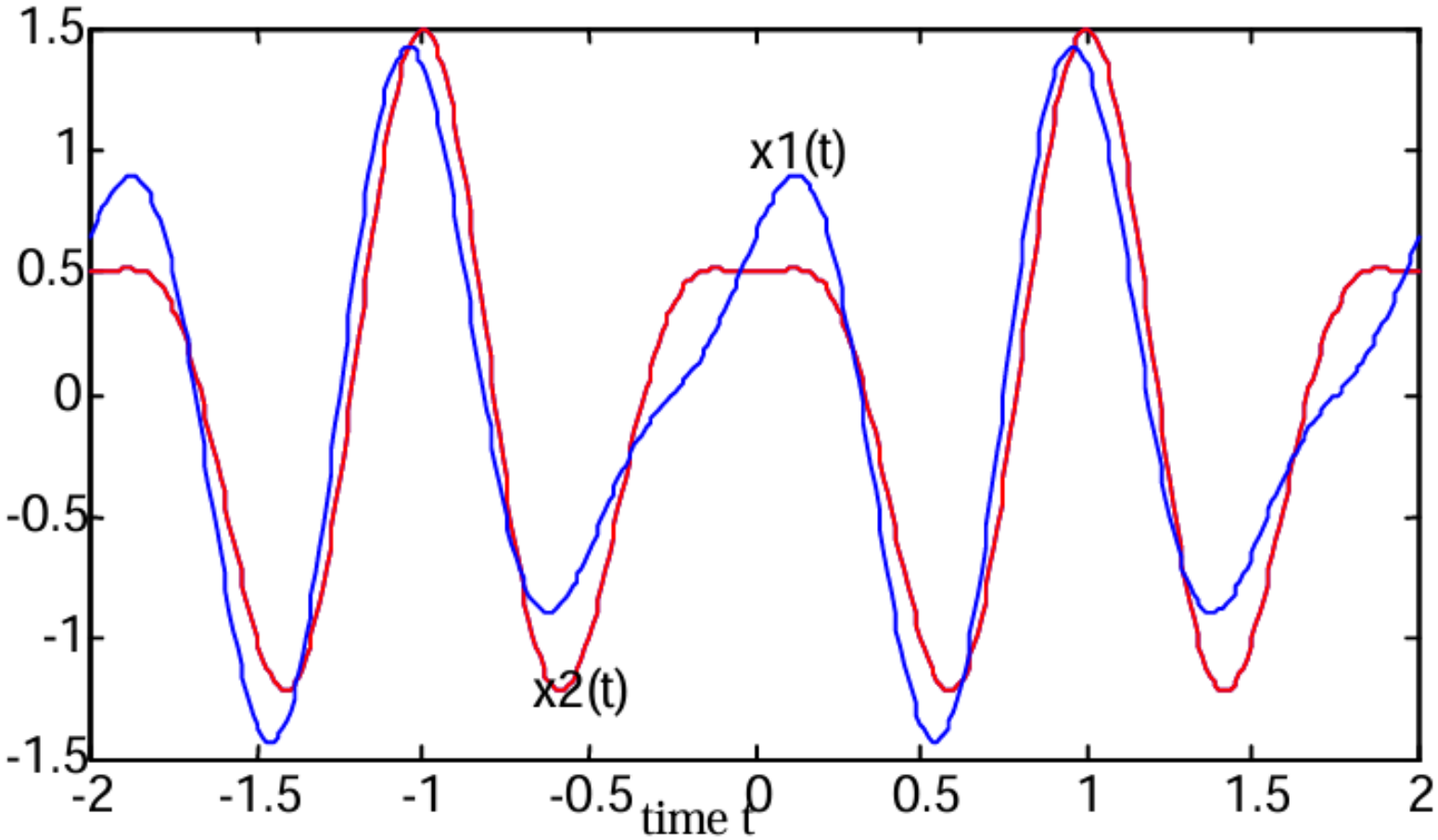
and

$$\begin{aligned} X_2(t) &= \cos(2\pi 2 f_0 t) - 0.5 cos(2\pi 3 f_0 t) \\ &= \cos(2\pi 2 f_0 t) + 0.5\cos(2\pi 3 f_0 t + \pi) \end{aligned}$$

with $f_0 = 0.5$ Hz.

Cont. . .

Thus each waveform is a combination of 1 Hz and 1.5 Hz cosine terms, with the same amplitudes (1 and 0.5) for each, but with the phase of the second cosine differing by $\frac{\pi}{4}$ in the two waveforms. The waveforms are significantly different from each other.



Cont. . .

❃ For any periodic waveform, the **Fourier spectrum** is the set of A_n coefficients together with their respective phases ϕ_n, where the index $n = 0, 1, 2, \ldots$ of course corresponds to frequency nf_0. Note that the dc term a_0 is the average value of the periodic function, and may be considered to be the zero-frequency ($n = 0$) term; we define $A_0 = |a_0|$ (there is no b_0 term).

❃ The **amplitude spectrum** refers only to the amplitudes A_n. It may be plotted as a function of n. Similarly, the **phase spectrum** is the phase ϕ_n as a function of n.

❃ The **frequency spectrum** comprises an amplitude spectrum, which is a graph of the amplitudes of each of the constituent sinusoidal components in the Fourier series plotted against frequency, and a phase spectrum which is their phases.

Example

Determine the frequency spectrum of the rectangular waveform with $a_0 = 1$, $a_n = 0$ and $b_n = \dfrac{1 - \cos n\pi}{n\pi}$.

Solution

We have $b_1 = \frac{2}{\pi} = 0.64$, $b_2 = 0$, $b_3 = \frac{2}{3\pi} = 0.21$, $b_4 = 0$, $b_5 = \frac{2}{5\pi} = 0.13$, etc. The A_0 term is 1. Using equation (24), the A_1 term is 0.64, the A_2 term 0, the A_3 term 0.21, the A_4 term 0, the $_5$ term 0.13, etc. The phases, when referred to a sine wave, are 0° for all components. When referred to a cosine wave they are -90°. Figure 11(a) shows the resulting amplitude spectrum and Figure 11(b) the phase spectrum when referring to sinusoidal components.

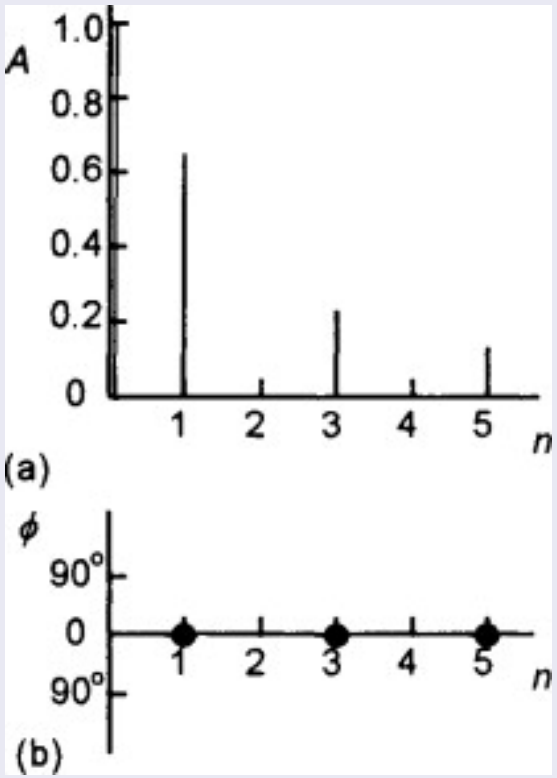


Figure 11: Frequency Spectrum

Example

Determine the frequency spectrum for a half-wave rectified sinusoidal waveform if it has the Fourier series:

$$y = \frac{1}{\pi} - \frac{2}{3\pi}\cos 2\omega t - \frac{2}{15\pi}\cos 4\omega t + \cdots + \frac{1}{2}\sin \omega t$$

Solution

The A_0 term is $\frac{1}{\pi} = 0.32$. Using equation (24), the A_1 term is 0.5, the A_2 term $\frac{2}{3\pi} = 0.21$, the A_3 term 0 and the A_4 term $\frac{2}{15\pi} = 0.04$. The phases, when referred to a sine wave are $\phi_1 = 0$ and since $-\cos\omega t = \sin(\omega t - 90^\circ)$, $\phi_2 = -90^\circ$ and $\phi_4 = -90^\circ$. Figure 12(a) shows the amplitude spectrum and Figure 12(b) the phase spectrum when referring to sinusoidal components.

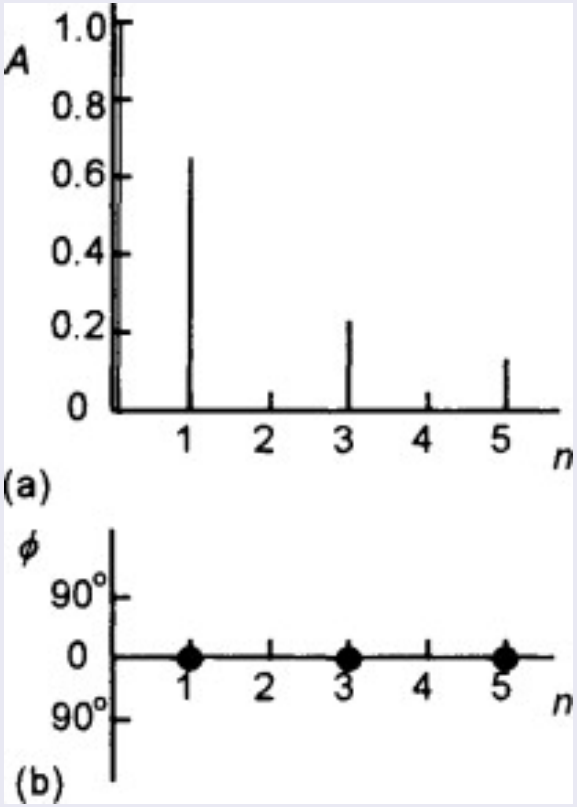


Figure 12: Frequency Spectrum